QDW90A Pressure Transmitter

I. Product Definition

The QDW90A Diffused Silicon Pressure Transmitter is an industrial-grade, high-precision pressure measurement and transmission device developed by Anhui Qidian Automation Technology Co., Ltd. Based on diffused silicon piezoresistive sensing technology, it linearly converts the pressure signal of the measured medium (liquid, gas, or steam) into a standard industrial electrical signal (4–20 mA). With stable signal output and strong anti-interference capabilities, it serves as a core component for pressure monitoring, control, and protection in industrial automation control systems.

Designed specifically for complex industrial field conditions, this product is suitable for a wide range of industries, including water treatment, petrochemicals, HVAC, hydraulics and pneumatics, and mechanical equipment. It supports continuous pressure measurement across a wide range from low pressure (0–0.1 MPa) to high pressure, and operates on a wide voltage supply range of 12–36 V, meeting the signal acquisition requirements of the vast majority of industrial PLC and DCS systems.

II. Working Principle

The operational process of the QDW90A diffused silicon pressure transmitter can be divided into four core stages: pressure signal sensing, signal conversion, signal amplification and conditioning, and standard signal output. A simplified explanation is provided below:

1. Pressure Signal Sensing Stage

The pressure of the measured medium is transmitted through the stainless steel isolation diaphragm at the front end of the transmitter and acts on the internal diffused silicon piezoresistive chip. Four piezoresistors are etched onto the surface of the diffused silicon chip, forming a Wheatstone bridge structure, which is the core component of pressure sensing.

2. Signal Conversion Stage

When pressure acts on the chip, the crystal lattice of the silicon deforms, causing the resistance of the pressure-sensitive resistors to change linearly with the magnitude of the pressure. The originally balanced Wheatstone bridge becomes unbalanced, outputting a weak voltage signal (millivolt level) proportional to the pressure, thereby completing the physical conversion from “pressure signal to electrical signal.”

3. Signal Amplification and Conditioning Stage

The millivolt-level raw electrical signal is processed through internal high-precision amplification circuits, temperature compensation circuits, and linearization conditioning circuits. This eliminates the effects of temperature drift and nonlinear errors, calibrating the signal into a stable output that maintains a strict linear relationship with pressure, while also adapting to the electromagnetic interference environment typical of industrial settings.

4. Standard Signal Output Stage

The conditioned signal is ultimately converted into a standard 4–20 mA current output widely used in industrial applications, where 4 mA corresponds to the lower limit of the measurement range (e.g., 0 MPa) and 20 mA corresponds to the upper limit (e.g., 0.1 MPa). This provides a signal output that is “transmissible over long distances, highly resistant to interference, and compatible with industrial control systems,” enabling data acquisition and processing by devices such as PLCs, DCS systems, and digital display instruments.

III. Core Value and Solutions to Pain Points

3.1 Overview of Core Value

1. High-Precision, Stable Measurement: With a measurement accuracy of 0.2% F.S., combined with temperature compensation and linearization design, it ensures precise and reliable pressure data across the entire range, preventing production control deviations caused by measurement errors.

2. Wide Range of Operating Conditions: The stainless steel housing and G1/4 standard threaded interface are compatible with the vast majority of industrial piping installation scenarios. It supports various media, including liquids, gases, and steam, meeting the operational requirements of different industries.

3. Industrial-Grade Interference Resistance: The 4–20 mA current signal transmission method, combined with industrial-grade circuit protection, effectively resists on-site electromagnetic interference, ensuring distortion-free long-distance signal transmission in complex industrial environments.

4. Wide Voltage Compatibility: The 12–36 V DC wide-voltage power supply is compatible with the power systems of most industrial sites, eliminating the need for dedicated power supplies and reducing system integration costs.

5. High Reliability and Long Service Life: The fully sealed stainless steel design meets industrial-grade protection standards, minimizing the impact of media corrosion, dust, and humidity on the equipment, thereby reducing maintenance frequency and replacement costs.

3.2 Key Challenges and Solutions

1. What should be done when inaccurate pressure measurements in industrial environments lead to production control errors?

Challenge: Traditional pressure sensors suffer from low accuracy and significant temperature drift. When on-site operating temperatures fluctuate or pressure varies, measurement data deviates significantly, leading to system misjudgments, loss of control over process parameters, and even production accidents.

Solution: The QDW90A employs diffused silicon piezoresistive sensing technology, combined with high-precision calibration to 0.2% F.S. and a built-in temperature compensation circuit. It maintains stable measurement accuracy across a wide temperature range with extremely low full-scale linear error, ensuring precise and reliable pressure data to provide accurate basis for production control.

2. How to address unstable pressure signal transmission caused by strong electromagnetic interference in industrial environments?

Challenge: Electromagnetic interference generated by equipment such as variable frequency drives and motors within factories can cause distortion in voltage signal transmission, leading to data jumps, signal loss, and other issues that affect the stable operation of the system.

Solution: The QDW90A employs a standard 4–20 mA current signal output. Current signals inherently possess anti-interference capabilities and are less susceptible to electromagnetic interference during transmission. Additionally, the device features a built-in anti-interference circuit, further enhancing signal stability and ensuring distortion-free data transmission over long distances.

3. Industrial environments are complex; how can we prevent sensor damage caused by media corrosion and moisture?

Challenge: Traditional sensors have low IP ratings, making them vulnerable to moisture, dust, and corrosive media, which leads to frequent equipment failures. Frequent replacements increase maintenance costs.

Solution: The QDW90A features a fully sealed stainless steel housing designed to meet industrial-grade protection requirements. It withstands moisture, dust, and non-aggressive corrosive media. Additionally, the stainless steel material offers excellent compatibility with various media, extending the device’s service life and reducing maintenance frequency and costs.

4. What should be done when on-site power systems are complex and sensor power supply compatibility is poor?

Challenge: Power specifications vary across different industrial sites, and some sensors have narrow operating voltage ranges, requiring the configuration of dedicated power supplies, which increases system integration costs and installation complexity.

Solution: The QDW90A supports a wide DC voltage range of 12–36 V, making it compatible with the vast majority of industrial field power systems. This eliminates the need for dedicated power supplies, simplifies the system integration process, and reduces equipment procurement and installation costs.

5. What if the sensor cannot be adapted to on-site conditions due to mismatched pipe installation interfaces?

Challenge: Some pressure sensors use non-standard threaded connections

that do not fit on-site piping, requiring the fabrication of custom adapters. This makes installation cumbersome and increases the risk of leaks.
Solution: The QDW90A features a standard G1/4 threaded connection, a universal specification for industrial piping. It fits directly onto the vast majority of industrial piping installations without the need for custom adapters, ensuring easy installation. Additionally, the threaded connection provides excellent sealing performance, effectively preventing media leaks.

IV. Key Parameters & Specifications

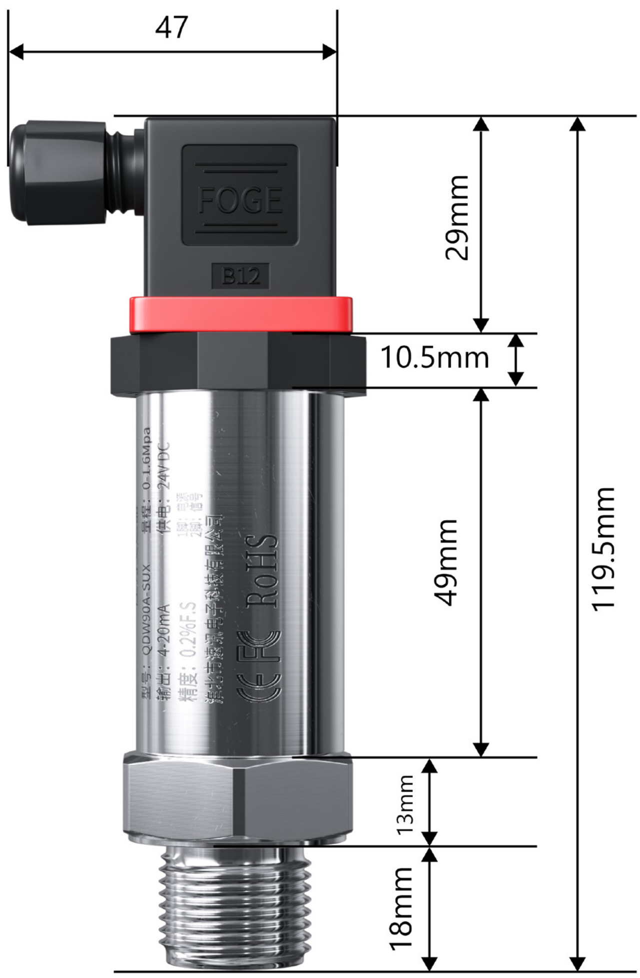
4.1 Basic Electrical Parameters
Sensor Core: Diffused silicon piezoresistive pressure sensor
Measured Media: Liquids, gases, steam (media compatible with 316L stainless steel)
Measurement Range: Standard range 0–0.1 MPa; other ranges available upon request (e.g., 0–0.6 MPa, 0–1.0 MPa, 0–2.5 MPa, 0–10 MPa, etc.)
Output Signal: Standard 4–20 mA DC current signal (two-wire)
Measurement Accuracy: ±0.2% F.S. (Full-scale accuracy)
Non-linearity Error: ≤±0.1% F.S.
Hysteresis Error: ≤±0.1% F.S.
Repeatability Error: ≤±0.05% F.S.
Temperature Drift: ≤±0.02% F.S./°C (within operating temperature range)
Zero error: ≤±0.1% F.S.
Full-scale error: ≤±0.1% F.S.
Response time: ≤10 ms
Insulation resistance: ≥100 MΩ (500 V DC)

4.2 Isolation Performance Parameters
Input/Output isolation: No built-in isolation (if isolation is required, an external signal isolator can be used)
Housing-to-Circuit Insulation Strength: 500 V AC for 1 minute, no breakdown or arcing
Common-Mode Rejection Ratio (CMRR): ≥80 dB
Differential-Mode Rejection Ratio (DMRR): ≥60 dB
Electromagnetic Interference (EMI) Immunity: Complies with CE/EMC standards; resistant to electromagnetic interference generated by industrial equipment such as inverters and motors
Signal Transmission Immunity: 4–20 mA current signal transmission effectively resists line interference, with a transmission distance of up to 1,000 meters (without repeaters)

4.3 Power Supply & Power Consumption Parameters
Supply Voltage Range: 12–36 V DC (typical value 24 V DC, industrial standard power supply)
Operating Voltage: 24 V DC (recommended, compatible with most PLC/DCS systems)
Reverse Polarity Protection: Supports reverse polarity protection; brief reverse connections will not damage the device
Full-Load Power Consumption: ≤0.5W (at 24V DC supply, maximum operating current 20mA)
Power Distribution Output: No additional power distribution output; this is a two-wire passive signal output that does not require a separate signal power supply
Ripple Suppression: ≤±0.1% F.S. (when power supply ripple is ≤10%)

4.4 Environmental Adaptation Parameters
Operating Temperature Range: -20°C to +80°C (medium temperature must not exceed this range; a heat-dissipating structure can be customized for high-temperature medium measurement)
Medium Temperature Range: -20°C to +125°C (short-term contact is supported; long-term operation must be maintained within the operating temperature range)
Storage Temperature Range: -40°C to +100°C
Operating Humidity Range: ≤95% RH (non-condensing)
Protection Rating: IP65 (standard configuration; dustproof and splash-proof, suitable for humid and dusty industrial environments)
Explosion-Proof Rating: Standard models do not have explosion-proof certification; explosion-proof versions can be customized for intrinsic safety applications, compatible with intrinsic safety barriers (e.g., GS8000 series, MTL series safety barriers)
Vibration Resistance: 10–2000 Hz, 10g, 2 hours in each of the three axes, with no mechanical damage or performance drift
Shock Resistance: 100g, 11 ms, half-sine wave, 3 times in each of the three axes, with no mechanical damage or performance drift

4.5. Physical Parameters
Mounting Method: Direct threaded mounting, suitable for direct installation on industrial piping
Connection Specifications: G1/4 male thread (standard configuration; other thread specifications such as M20×1.5 and G1/2 are available)
Housing Material: Main body made of 304 stainless steel (316L stainless steel available for corrosive media); electrical connector made of engineering plastic (black, IP65 protection rating)
Seal Material: Fluorocarbon rubber (FKM, standard configuration, suitable for general media; optional nitrile rubber NBR, silicone rubber VMQ, etc.)
Dimensions:
Overall Length: Approx. 118.5 mm (including electrical connector)
Body Diameter: Approx. 30 mm (stainless steel cylindrical section)
Mounting Thread Length: Approx. 15 mm
Hexagon Face-to-Face Dimension: 24 mm (compatible with standard wrenches)
Weight: Approx. 200 g (standard configuration, excluding mounting hardware)
Indicator Light Definition: Standard models do not include an indicator light (if operational status indication is required, a version with an indicator light can be customized; indicator light definition: power light remains on for normal power supply, flashing indicates a fault or signal anomaly)
Electrical Connector Type: Hirschmann-type waterproof connector, standard configuration, compatible with industrial cables, supports quick plug-and-play
Protection Details: Fully sealed, welded stainless steel body; electrical connector features an integrated waterproof O-ring; threaded interface includes a built-in sealing groove; can be paired with PTFE tape or sealing gaskets to achieve an IP65 protection rating, effectively preventing dust, water, and humid air from entering the device interior.



V. Application-Specific Selection Guide

5.1 Key Selection Criteria

1. Signal Type Selection: For standard industrial automation, PLC, and DCS systems, the QDW90A standard 4–20 mA two-wire current output version is recommended as the first choice due to its strong noise immunity, long transmission distance, and highest versatility. If digital communication or remote networking is required on-site, please confirm in advance to order a customized digital signal output version.

2. Channel Selection: The QDW90A features a single-channel, independent pressure measurement design. Each unit corresponds to a single pressure measurement point; therefore, a separate transmitter must be configured for each monitoring point on-site. It does not have built-in multi-channel multiplexing functionality; multiple measurement points require multiple units connected in parallel.
Power Supply Selection: For standard field applications, select the 12–36 VDC wide-voltage version; the industrial standard of 24 VDC ensures stable operation. In environments without a dedicated regulated power supply or where voltage fluctuations are significant, prioritize the wide-voltage version to avoid the need for an additional voltage regulator module.

3. Selection Based on Explosion-Proof Requirements: For general indoor applications, standard industrial facilities without flammable or explosive gases or dust, water treatment, and HVAC environments, the standard QDW90A general-purpose model is sufficient. For petrochemical, oil and gas storage and transportation, pharmaceutical dust workshops, and enclosed spaces with flammable or explosive atmospheres, the intrinsically safe (IS) custom model must be used; the standard model must not be substituted under any circumstances.

4. Selection Based on Installation Environment: For standard indoor environments that are dry and dust-free, the standard IP65-rated version can be used directly; for outdoor, open-air environments, or conditions involving moisture, rain, high dust levels, or corrosive water vapor, prioritize the version with a 316L stainless steel diaphragm and a fully sealed, waterproof structure; in environments with strong vibrations or intense interference from frequency converters, prioritize pairing with an external signal isolator to enhance stability.

5. Range Selection: Select a range equivalent to 1.2–1.5 times the actual on-site working pressure. Do not select a range that is too small, as this may cause damage due to overpressure; similarly, do not select a range that is too large, as this may reduce measurement accuracy in the low-pressure range.
Media Compatibility: For clean water, ordinary gases, and non-corrosive media, use the standard 304 stainless steel model; For acids, alkalis, mildly corrosive liquids, and chemical media, the version with a 316L stainless steel isolation diaphragm must be selected.

5.2. Selection Guidelines for Industry Applications

1. For pressure monitoring in conventional water systems, atmospheric-pressure fluids, and ordinary gases in the chemical industry, directly select the standard QDW90A 4–20 mA output version with a standard range; For environments involving flammable or explosive media or in enclosed chemical workshops, select the intrinsically safe explosion-proof custom model and use it in conjunction with a safety barrier.

2. For pressure monitoring in auxiliary control systems, circulating water, boiler auxiliary piping, and HVAC systems in the power industry, the standard QDW90A model with IP65 protection and 24VDC power supply is sufficient to meet long-term stable operation requirements.

3. For pressure measurement in circulating hydraulic systems, cooling water pipelines, and pneumatic control systems within the metallurgical industry—where on-site electromagnetic interference and vibration are significant—we recommend the standard QDW90A model with an external signal isolator to enhance resistance to interference and vibration.

4. For pressure measurement in cleanrooms, process fluids, and pure water pipelines in the pharmaceutical industry, the QDW90A custom version—featuring a 316L stainless steel diaphragm and a fully sealed, dead-space-free design—is suitable for clean environments and mildly corrosive media. For areas with flammable or explosive dust, the explosion-proof version should be selected.

5. For pressure monitoring in non-standard equipment, hydraulic and pneumatic systems, and assembly lines within the smart manufacturing and automation equipment industry, select the QDW90A standard two-wire 4-20mA output version. It features universal interfaces and convenient installation, making it compatible with various PLCs and industrial control instruments.

6. For pressure monitoring in wastewater, water purification, and water supply/drainage networks within the water treatment industry, use the standard version for conventional conditions; for corrosive wastewater media, select the 316L diaphragm upgrade version; for outdoor installation, choose the IP65 fully sealed protection configuration.

5.3 Common Selection Misconceptions

1. Misconception regarding isolation level: Many sites directly select standard transmitters, neglecting the electromagnetic interference caused by variable frequency drives and high-power motors, which leads to signal fluctuations and value drift; failure to implement isolation measures in high-interference environments results in difficult commissioning and unstable system operation.

2. Misjudging Signal Range: Selecting a range based solely on rated working pressure without allowing for a safety margin can lead to sensor damage when operating conditions fluctuate and exceed pressure limits; conversely, blindly selecting a wide range results in poor resolution at low pressures and a significant drop in control accuracy.

3. Misconception regarding load matching: Failing to match the input load resistance of downstream PLCs or instruments, exceeding the transmitter's maximum load capacity, results in signal attenuation and inaccurate output, leading to issues such as 4 mA zero-point offset and 20 mA full-scale values failing to meet standard requirements.

4. Misapplication of General-Purpose Models in Explosion-Proof Environments: In hazardous areas involving flammable and explosive chemicals, oil and gas, or pharmaceutical dust, arbitrarily selecting standard non-explosion-proof QDW90A models poses safety risks, fails to meet safety regulations, and will result in failure during final acceptance inspections.

5. Misconception regarding neglecting material compatibility: Using the standard 304 stainless steel version for corrosive media leads to long-term diaphragm corrosion, leakage, and measurement failure, shortening the equipment's service life.

6. Misconception regarding arbitrary selection of interface specifications: Failing to verify the on-site pipe thread specifications and blindly assuming a standard G1/4 interface results in installation issues upon delivery, requiring the addition of adapter fittings, which increases the risk of leakage and construction costs.

5.4 Customization Recommendations

1. Custom Signal Formats: For on-site requirements such as 0-5V or 0-10V voltage output, RS485 digital communication, or Modbus protocol, the QDW90A supports programmatic and circuit customization to meet the integration needs of non-standard industrial control systems.

2. Customization for Non-Standard Operating Conditions: For high-temperature media, extremely cold outdoor environments, highly corrosive media, and high-vibration/shock conditions, we can customize high-temperature heat dissipation structures, low-temperature/wide-temperature versions, 316L/Hastelloy special diaphragms, and reinforced anti-vibration structures. Additionally, we can customize non-standard threaded connections, extended pressure ports, and special sealing materials.

3. Multi-Point Monitoring Customization: For on-site scenarios requiring centralized monitoring of multiple channels and pressure points, we offer a turnkey solution consisting of multiple QDW90A transmitters bundled with an external multi-channel signal isolation module, eliminating the need to modify individual transmitters. This approach enables unified cabling and integration with the control system, simplifying installation,

commissioning, and future maintenance.

4. Customized Features: We support personalized customization options such as on-site zero-point adjustment, status indicator lights, lightning surge protection, and intrinsically safe explosion-proof certification to meet the safety regulations and operational requirements of specific industries.

VI. Installation and Commissioning Guidelines

6.1. Pre-installation Preparation

1. Checking Accessories

Upon delivery, first take inventory of the standard accessories, including the main unit, waterproof electrical connectors, sealing gaskets, spare installation gaskets, and user manuals. Verify that the product's range, output signal, threaded connection specifications, and housing material match the selected model. Ensure there are no defects such as external dents, damaged threads, deformed diaphragms, or oxidized terminals. Do not install the unit if any accessories are missing or do not match the specified model.

2. Environmental Inspection

At the installation site, ensure there are no persistent strong corrosive gases or intense open flames and high-temperature heat sources. Avoid areas with strong electromagnetic radiation, such as those near variable frequency drives, high-power motors, and high-voltage cables. For outdoor installations, ensure basic conditions for protection against rain, dust, and direct sunlight are met. Ensure that the pipeline measurement point is free from severe mechanical impact and long-term high-frequency vibration, and that the medium does not experience sudden overpressure surges. If environmental conditions are not met, install buffer devices, vibration-damping brackets, or protective enclosures in advance.

3. Tool List

Prepare an adjustable wrench, Allen wrenches, insulated screwdrivers, wire strippers, a multimeter, standard industrial shielded cable, PTFE tape, electrical tape, and cable ties. If on-site calibration is required, have a standard pressure calibrator and a DC regulated power supply on standby.

4. Safety Precautions

Disconnect the main power supply to the on-site equipment before installation. Under no circumstances should wiring be performed or electrical connectors be plugged in or unplugged while the system is energized. Disassembly and installation work may only be performed after the pressure pipeline has been fully depressurized; it is strictly prohibited to disassemble the transmitter while it is under pressure. During wiring, ensure that bare wires are not exposed and that there are no misconnections or short circuits. In flammable and explosive areas, explosion-proof construction regulations must be strictly followed; for standard models, opening the cover to perform wiring in hazardous areas is strictly prohibited.

6.2 Standard Installation Steps

1. Mounting and Securing the Transmitter Body

The QDW90A features a threaded direct-mount design and does not require a mounting rail. Clean debris from the pipe connection threads, apply an appropriate amount of PTFE tape, and manually screw the unit into the connection. Use a wrench to tighten it moderately; do not force it, as this may cause the threads to strip or damage the internal diaphragm due to excessive pressure. Install the unit vertically or horizontally in the correct orientation, with the pressure port facing downward or to the side to prevent debris from accumulating and blocking the pressure measurement orifice.

2. Wiring Specifications

This product is a two-wire 4-20mA universal type. It distinguishes only between power supply and signal lines, which share the same positive and negative terminals; there are no separate input terminals.

Connect the positive terminal to the positive supply line of the 24 V DC power source; this terminal also serves as the positive terminal for the 4–20 mA signal output.

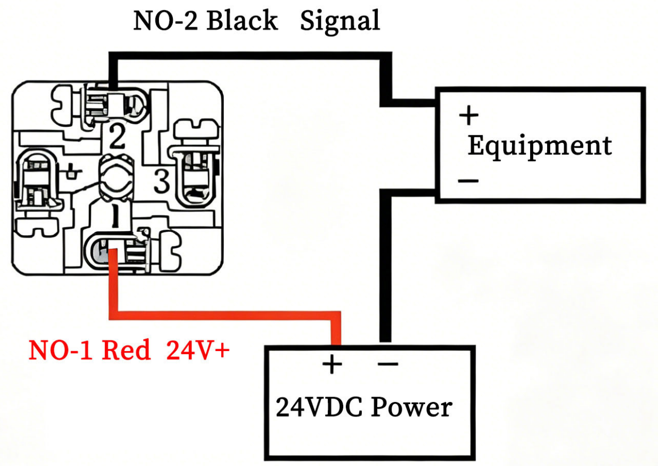
Connect the negative terminal to the negative supply line of the power source; this terminal also serves as the negative return line for the signal.

Use shielded twisted-pair cable throughout the installation. Route the cable away from power cables and high-voltage cables; do not install it in the same conduit as high-voltage power lines.

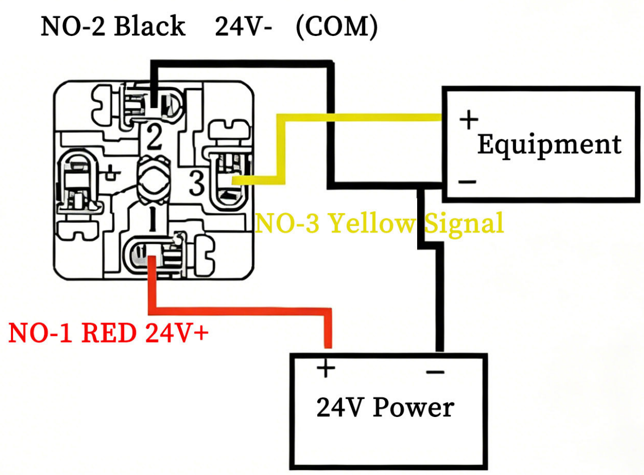
3. Grounding Requirements

The transmitter housing, on-site metal piping, and control cabinet must be reliably grounded; the cable shield must be grounded at a single point, specifically on the control cabinet side only. The shield at the transmitter end must remain floating (ungrounded) to prevent multi-point ground loops that could cause signal interference or value fluctuations.

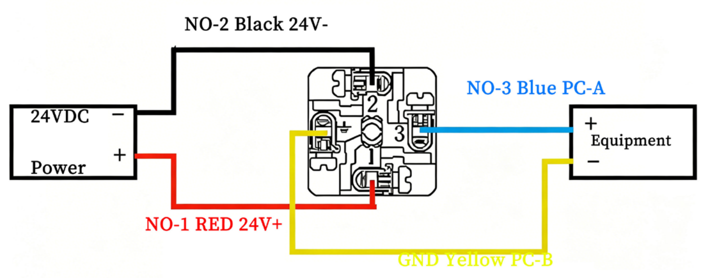
6.3. Textual Specifications for Wiring Diagram



4-20mA 2-Wire Current Signal Output



3-Wire Voltage Signal Output



RS485 Output 4-Wire

Product Structure

6.4 Power-On Commissioning Procedure
1. No-Load Test
After completing the installation and wiring, double-check the circuit for any misconnections or short circuits, and confirm that the pipeline is in a pressure-free, no-load state. Slowly apply the 24V operating power supply; the device should power on normally without overheating or unusual noises, and enter standby mode.

2. Signal Verification
Connect a multimeter set to the current range in series with the circuit. Under no-load conditions, the standard output should be close to the 4 mA zero-point reference value. Slowly pressurize the pipeline to the rated working pressure and observe the current rising linearly to the corresponding value; at full scale, it should be close to 20 mA. The entire process should be smooth, with no sudden jumps or stuttering.

3. Status Determination
The standard QDW90A model has no indicator lights; rely on the multimeter readings and the industrial control system display. For custom versions with indicator lights: the power light remains on when power is normal; the signal light remains steady without flashing when the signal is normal and stable; the indicator lights flash frequently when values are abnormal or there is a circuit fault.

4. Calibration Methods
For zero-point deviation, perform software-based zero-point fine-tuning via a downstream secondary instrument or PLC system; under no circumstances should the transmitter housing be opened to adjust internal potentiometers. If full-scale linear deviation is excessive, compare readings using a standard pressure calibrator and have the unit factory-calibrated; on-site hardware adjustments are not recommended.

6.5 Installation Precautions
1. Never operate the device beyond its rated pressure or full-scale range. Avoid sudden pressure surges in the medium that far exceed the product's rated range, as this may cause permanent damage to the diffused silicon chip and result in measurement failure.

2. Never exceed the rated supply voltage; do not exceed the 36 V DC upper limit. Do not connect to 220 V AC power to prevent direct burnout of internal circuits.

3. Avoid routing cables in close proximity to variable frequency drives, motors, or high-voltage power lines; do not share cable trays to minimize electromagnetic coupling interference.
In damp, rainy, or underground environments, ensure waterproof connectors are securely tightened and cables are properly sealed. Do not allow electrical connectors to be exposed to rain to prevent water ingress, short circuits, and corrosion of terminals.

4. Do not use excessive force with a wrench to over-tighten threads, and do not strike the transmitter body. This prevents mechanical stress from causing permanent zero-point drift in the internal sensor chip.

5. In corrosive media applications, do not use the standard 304 diaphragm version directly. Do not install it without proper material matching, as this may lead to diaphragm corrosion and leakage later on.

VII. Recommendations for Daily Use & Operation and Maintenance
7.1 Key Points for Daily Inspections
1. Visual Inspection
Conduct daily visual inspections of the transmitter housing to check for dents, deformation, or corrosion. Inspect threaded connections for media leaks, scaling, or rust. Ensure that electrical waterproof connectors are securely fastened with no loosening or detachment, and that the housing seals show no cracks or signs of water ingress or condensation.

2. Signal Stability Monitoring
Periodically check the pressure readings displayed on the PLC, DCS, or secondary instruments to observe for any irregularities such as erratic fluctuations, zero-point drift, value freezing, or sudden spikes or drops. Compare these readings with the actual on-site process pressure to confirm that the measurement data responds normally and shows no lag or deviation.

3. Environmental Temperature and Humidity Monitoring
Inspect the ambient temperature around the installation to ensure it does not exceed the product's normal operating range, avoiding prolonged exposure to high temperatures or freezing conditions. When on-site air humidity is excessively high or moisture is heavy, focus on checking the electrical junction box for condensation or moisture to prevent signal abnormalities caused by reduced insulation.

4. Operating Load Inspection
Monitor for instantaneous pressure surges or water hammer in the pipeline medium to avoid prolonged pressure exceeding the rated range; In areas near equipment subject to strong vibrations, observe whether the transmitter resonates with the equipment to prevent long-term vibration from causing internal structural loosening or accuracy drift.

5. Wiring Condition Inspection
Visually inspect external cables for signs of aging, fraying, or excessive bending. Ensure wiring is intact and free from potential hazards such as compression by heavy objects, abrasion, or rodent damage.

7.2. Regular Maintenance Plan
1. Cleaning and Maintenance Schedule
In standard industrial environments, perform surface cleaning every 3 months; in environments with heavy dust, high humidity, or chemical corrosion, cleaning is recommended monthly. Use a dry, soft cloth to wipe dust, oil, and crystallized debris from the housing and connector surfaces. Do not directly rinse electrical connectors with water, and do not use strong acid or alkali solvents to wipe the unit.

2. Wiring Tightness Inspection
Conduct a terminal inspection every six months. After powering down, verify that internal wiring connections are secure, free of loose or oxidized contacts, and show no signs of blackening. Ensure the shield grounding terminal is firmly secured with no loosening or disconnection to prevent signal interference caused by poor contact.

3. Recommendations for Accuracy Recalibration
Under normal operating conditions, perform accuracy calibration once every 12 months. In high-demand scenarios such as chemical processing, pharmaceuticals, and precision process control, recalibrate every 6 months. Use a standard pressure calibrator to compare measured values with instrument readings, and record zero-point and full-scale linearity errors. If deviations exceed the allowable range, perform a system zero-point fine-tuning or return the instrument to the factory for recalibration.

4. Seal and Connection Inspection
Inspect pressure threaded connections and seal gaskets annually for signs of aging. Replace gaskets immediately if they are hardened, deformed, or leaking. Re-seal threaded areas to prevent slow media leakage that could affect operating conditions and equipment lifespan.

5. Grounding System Re-inspection
Annually verify the integrity of the instrument housing grounding, single-point shield grounding, and control cabinet grounding. Ensure grounding resistance remains within specified limits to prevent interference drift caused by grounding failure or multi-point grounding.

7.3 Long-Term Maintenance Tips
1. Long-Term Dust and Moisture Protection
Whether the equipment is idle for extended periods or in continuous operation, ensure that all electrical waterproof connectors remain fully locked and sealed at all times. In outdoor, underground, or humid workshop environments, install additional protective rainproof enclosures and dust covers to prevent dust accumulation, rain splashes, and internal condensation caused by diurnal temperature fluctuations. When storing idle equipment, place it in a dry, well-ventilated indoor area to avoid outdoor stacking, moisture absorption, and mold growth.

2. Optimized Maintenance for Interference Resistance
Use shielded twisted-pair cables for all signal lines. Keep them away from variable frequency drives, high-power motors, and high-voltage power cables; do not route them through the same conduits or cable trays as high-voltage power lines. In locations where slight signal fluctuations have already occurred, install external signal isolators and surge protection modules to provide long-term, stable suppression of electromagnetic interference.

3. Protection Techniques for Extreme Environments
In high-temperature environments, avoid direct exposure to open flames or high-temperature radiation from pipes; install heat-dissipating buffer tubes if necessary. For low-temperature and severe cold outdoor conditions, select a wide-temperature range version and apply external thermal insulation to prevent measurement drift caused by low temperatures. For long-term use with corrosive media, prioritize regular inspection of the diaphragm's corrosion status; upgrade to the 316L stainless steel diaphragm version if necessary.

4. Long-Term Protection of Installation Structure
In high-vibration environments, install vibration-damping brackets and cushioning mounts to reduce the impact of mechanical resonance on the transmitter's internal chips; avoid striking or prying the housing with external force; during routine inspections, prevent accidental collisions and ensure the installation remains stable without deformation from external forces.

5. Long-Term Protection Through Proper Power Usage
Strictly maintain the supply voltage within the 12–36 VDC range to avoid sudden voltage fluctuations and excessive power supply ripple. Install voltage regulation and filtering modules at the power input to prevent damage to internal circuits caused by power grid fluctuations and surge impacts, thereby extending the service life of the entire unit.

VIII. Troubleshooting & Emergency Response
8.1 List of Common Faults
The transmitter has no output signal; the downstream instrument or PLC displays no values
Signal zero drift; values gradually deviate; no-load current does not meet the 4 mA reference
Signal does not track pressure changes; readings remain frozen
Measured values exhibit erratic fluctuations, high volatility, or interference-induced distortion
Operating conditions are normal, but readings are abnormally high or low, indicating overall accuracy deviation
Abnormal indicator lights on custom models: constantly off, constantly on, or frequent flashing alarms
System alarms are triggered inexplicably or falsely under normal pipeline pressure
Operates normally for a short time, but signal is interrupted or becomes corrupted after a period of operation

8.2 Analysis of Failure Causes
1. Wiring-related causes: Reversed polarity, loose or faulty terminal connections, shielding wire grounded at both ends, signal and power lines routed in the same conduit, or exposed wiring causing short circuits.

2. Power supply-related causes: Supply voltage below 12 V DC or exceeding 36 V DC, excessive power supply ripple, regulated power supply failure, or circuit load resistance exceeding limits.

3. Signal-related causes: Mismatched load on the downstream instrument, incorrect range setting, excessive number of devices in series on the circuit, or lack of anti-interference measures for long-distance signal transmission.

4. Environmental interference causes: Strong electromagnetic radiation from nearby variable frequency drives, high-power motors, or high-voltage cables; moisture condensation or dust entering the terminal compartment leading to reduced insulation; high vibration or resonance causing internal parameter drift.

5. Operational Conditions: Long-term overpressure operation; water hammer damage to the diffused silicon chip; medium crystallization blocking the pressure measurement orifices; corrosion of the diaphragm by corrosive media leading to measurement failure.

6. Device-Related Causes: Aging internal circuits; sensor chip drift; water ingress due to seal failure; deviation from factory calibration parameters; hardware damage.

8.3 Step-by-Step Troubleshooting
1. No Output Signal, No Display on Backend
Troubleshooting Steps:
First, disconnect power and check whether the wiring is reversed (positive and negative terminals swapped) or terminals are loose; use a multimeter to measure whether the supply voltage is within the normal range of 12–36 V DC; check whether signal wires are broken, shorted, or have exposed insulation causing ground loops; confirm that the current acquisition port mode on the backend PLC or instrument is set to 4–20 mA input.
Remedial Actions:
Re-crimp terminals to correct polarity; replace damaged cables; restore standard 24 V DC power supply; correctly configure the input type on the backend instrument; clear short-circuit points in the wiring.
Emergency Workaround:
If production must be restored immediately on-site, temporarily replace the transmitter with a spare QDW90A unit of the same range and signal specifications to restore system operation; repair the faulty unit later.

2. Signal Zero Drift and Inaccurate No-Load Readings
Troubleshooting Steps:
Confirm that the pipeline is fully depressurized and unloaded, and observe whether the output deviates from 4 mA; check for sudden changes in ambient temperature or excessive on-site vibration; verify whether the shielded cable is grounded at both ends, causing loop current interference.
Remedial Measures:
Perform a software zero-point fine-tuning via the backend PLC or secondary instrument; modify the shielding to single-ended grounding; reinforce the installation with vibration dampers to reduce the impact of mechanical vibration; avoid areas exposed to direct high heat or freezing temperatures.
Emergency Workaround:
Temporarily lock the backend reference zero point to maintain production; perform a unified calibration during a scheduled shutdown window or return the unit to the factory for calibration.

3. Pressure Fluctuations, Signal Lag, or Stuck Readings
Troubleshooting Steps:
Check whether the pressure measurement port is blocked by contaminants, scale, or crystallization; verify if the measurement range parameters have been accidentally modified; inspect the circuit for loose connections or broken wires.
Remedial Actions:
Disassemble and clean debris from the pressure port; remove scale from the diaphragm surface; restore the standard measurement range

parameters; Tighten all terminal blocks and replace aged cables.
Emergency Workaround:
Bypass the issue by using a backup measurement loop and temporarily switch monitoring points to avoid disrupting mainline production.

4. Value Jumps, Interference Distortion, and Erratic Fluctuations
Troubleshooting Steps:
Check whether signal cables are routed in the same cable tray as power cables or run parallel at close proximity; verify if variable frequency drives or high-frequency equipment are operating nearby; check for improper grounding or multiple grounding points.
Remedial Actions:
Route signal cables through separate conduits, away from high-voltage lines; ensure the shielding layer is reliably grounded at a single point; install external signal isolators to reduce interference; optimize cabinet grounding to lower ground resistance.
Emergency Workaround:
Increase digital filter parameters in the backend software to temporarily suppress value fluctuations and ensure process stability.

5. Overall Measurement Errors (High/Low) or Accuracy Deviations
Troubleshooting Steps:
Verify whether the selected measurement range matches actual operating conditions; check for prolonged overpressure or water hammer impacts; verify if ambient temperature and humidity exceed the operating range.
Remedial Actions:
Re-select an appropriate measurement range; install buffer pipes and shock absorbers to prevent sudden impacts; return the unit to the factory for re-calibration if the deviation exceeds the allowable range.
Emergency Workaround:
Set a fixed offset compensation value in the backend to temporarily correct the displayed values without affecting process control.

6. Abnormal Indicator Lights or False Alarms on Custom Models
Troubleshooting Steps:
Check for unstable power supply or loose connections; verify if signals are out of range or if there are circuit faults; check for circuit abnormalities caused by moisture ingress due to a humid environment.
Remedial Actions:
Ensure a stable power supply and retighten connections; verify if the process pressure is actually out of range; properly seal the terminal compartment to prevent moisture ingress.
Emergency Workaround:
Temporarily disregard the indicator light status and monitor production based on the actual pressure values displayed in the backend.

8.4. After-Sales Repair Guidelines
1. Criteria for Determining Malfunctions
The following situations indicate a malfunction in the device itself; it is recommended to report the issue directly for repair:
All wiring, power supply, and circuits have been checked and are normal, yet there is still no output or persistent severe drift;
No overpressure or improper installation, yet measurement accuracy remains consistently out of tolerance and cannot be calibrated on-site;
Water ingress, damaged housing, deformed diaphragm, or obvious physical damage;
Recurring signal interference or suspected internal circuit damage that cannot be repaired on-site.

2. Repair Process
Record the device model, measurement range, duration of installation and use, and the fault symptoms;
Conduct an on-site inspection of wiring, power supply, environment, and backend parameters, and keep a brief description on file;
Contact the manufacturer’s after-sales service, providing the model, fault symptoms, and operating conditions;
Return the faulty device according to the instructions and await testing, calibration, or repair/replacement.

3. Warranty Coverage
The QDW90A Diffused Silicon Pressure Transmitter is covered by standard warranty service provided it has been properly installed, operated under compliant conditions, and is free from human-caused damage or unauthorized disassembly or modification; impact damage, diaphragm corrosion due to misuse with corrosive media, or water ingress from exposure to rain without proper protection is not covered by the free warranty; however, paid repair and replacement services are available.

4. Precautions for Use
Do not attempt to open the housing or disassemble the internal circuitry and sensor chip of a malfunctioning device. Unauthorized disassembly will void the warranty and may result in the permanent loss of calibration parameters and complete device failure.